1(a) Charlie is about to study a computer games design course at university. He needs to purchase a computer to allow him to design and edit the graphics in his games, including video files.

Charlie's computer contains a data bus, an address bus and a control bus.

State the purpose of each of these three buses.

Data bus ____

____

Address bus ____

____

Control bus ____

____ [3]

(b) Charlie will need to use input **and** output devices when designing and editing his computer games.

i. State **one** input device **and one** output device that Charlie could use **and** give an example use of each.

Input device

Device name ____________________

____

Example use ____________________

____

Output device

Device name ____________________

____

Example use ____________________

____________________ **[4]**

ii. Charlie will need to store a large collection of files containing high definition images and high definition videos that he will use during his course.

State a type of secondary storage device that Charlie could use and give a reason why.

Type of storage ____________________

____

Reason ____________________

____________________ **[2]**

(c) In addition to the Central Processing Unit (CPU), Charlie will purchase a Graphical Processing Unit (GPU) that will be used to process and display the graphics in his games.

i. Tick (✓) **one** box on each row to indicate if each statement is referring to a CPU **or** a GPU.

Statement	CPU	GPU
Suited to performing simple operations in parallel on larger data sets		
Suited to performing complex operations on smaller data sets		

[2]

ii. State **two** benefits of using a GPU to process and display the graphics in computer games.

Benefit 1 ....

....

Benefit 2 ....

.... [2]

(d) Charlie will require an operating system that will allow him to have several applications open at the same time.

i. State which type of operating system Charlie will need to install.

[1]

ii. An operating system that Charlie is thinking about installing uses a multi-level feedback queue.

Describe how a multi-level feedback queue works.

[3]

iii. An operating system manages and installs device drivers.

Explain why hardware devices require a device driver.

[2]

(e) Once Charlie has installed an operating system he will need to install utility software.

State **two** utilities that Charlie could install and state the purpose of each.

Utility 1 ____

____

Purpose ____

____

Utility 2 ____

____

Purpose ____

____ **[4]**

2(a) OCR Consultants is designing a computer for a large company.

OCR Consultants is designing the architecture of the CPU.

i. Compare **two** differences between the Von Neumann architecture and the Harvard architecture.

Difference 1 ____

____

____

____

Difference 2 ____

____

____

____ **[4]**

ii. OCR Consultants is considering using pipelining.

State **two** benefits of using pipelining.

Benefit 1 ____________________

____

Benefit 2 ____________________

____________________ **[2]**

iii. OCR Consultants can base the CPU around a RISC architecture or a CISC architecture.

Compare **two** differences between a RISC architecture and a CISC architecture.

Difference 1 ____________________

____

____

____

Difference 2 ____________________

____

____

____________________ **[4]**

iv. Describe a benefit of making the CPU a multicore CPU.

____

____

____

____________________ **[2]**

(b) The employees of the large company work remotely from home and use large databases that store a lot of sensitive data. They need to be able to use the databases to search and sort data quickly.

OCR Consultants is considering two approaches:

- **thin client approach** – this is where the database is stored virtually on a remote server. All of the data processing is also completed on a remote server on virtual machines. The employees' computers are only used to capture the data input and display the results from the virtual machines
- **thick client approach** – this is where most of the processing is done on the employees' computer using specialised software to access the remote database.

Discuss the benefits and drawbacks of each of these approaches.

You should include the following in your answer:

- what is meant by virtual machines and virtual storage
- the benefits and drawbacks of a thin client and thick client approach described in the question
- a conclusion justifying which approach would be most suitable for this scenario.

[9]

3(a)

i. Convert the denary number **189** into an 8-bit binary number.

[1]

ii. Convert the denary number **189** into a hexadecimal value.

[1]

iii. Convert the hexadecimal value **E4** into a denary number.

[1]

(b) Explain the effect that a right shift of **three** places will have on a positive binary number.

[2]

(c) Show the subtraction of these **two** floating point binary numbers.

Both numbers are stored in a normalised floating point format, using 6 bits for the mantissa and 4 for exponent.

You should show your result in the same format.

Show your working out.

011010 0011 – 010010 0010

Working space

Final 6-bit **mantissa** ....

Final 4-bit **exponent** ....

[6]

(d)

i. Show the result of applying a bitwise **AND** mask of 1010 1110 to the byte 1010 0011.

Byte	**1010 0011**
AND Mask	**1010 1110**
Result	

[2]

ii. State the purpose of the bitwise **AND** mask.

--- **[1]**

4(a) Kofi is a software engineer and has been asked to write some code using the Little Man Computer (LMC) instruction set.

State the LMC command that would be used for each description given in the table.

Description	LMC Command
Load the value stored in the memory address into the accumulator	
Branch without checking the value in the accumulator	
Store the value from the accumulator into a memory address	
Branch if zero is in the accumulator	
Branch if zero or a positive number is in the accumulator	

[3]

(b) Kofi has been asked to write a program that inputs **two** numbers and then outputs the largest of the two numbers.

Write the code for this program using the LMC instruction set.

You can assume that the two numbers will always be different.

[4]

(c) Kofi is working on a number of projects. These include:

- **project 1** – Writing a mobile phone app to look at the news
- **project 2** – Writing software to send temperature readings from an embedded heat sensor to other devices
- **project 3** – Writing an image editing application for a desktop computer
- **project 4** – Writing software to calculate the results of a physics experiment that produces terabytes of data
- **project 5** – Writing a new device driver for a new piece of hardware.

Discuss which of these projects should be written in a high level programming language and which should be written in a low level language.

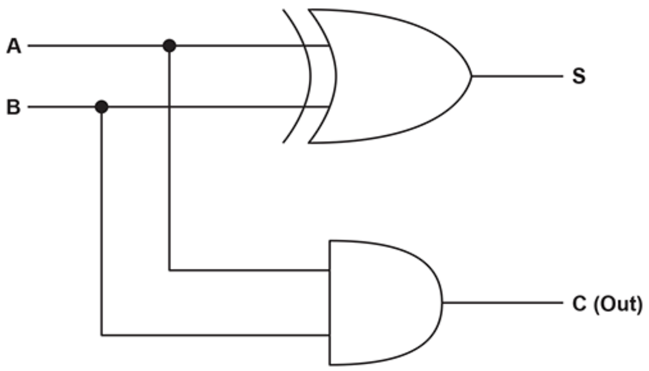
You should include the following in your answer:

- what is meant by high level programming language and low level language
- which projects are suitable for high level and which are suitable for low level languages

- a conclusion that justifies why these different languages are needed.

[12]

5(a) This diagram shows the logic circuit for a **half** adder.



Complete the truth table for this half adder circuit.

A	B	S	C (Out)
0	0		
0	1		
1	0		
1	1		

[4]

(b) Describe the difference between a half adder circuit and a full adder circuit.

---

[2]

(c) The diagram below shows a partially completed logic circuit for a **full** adder.

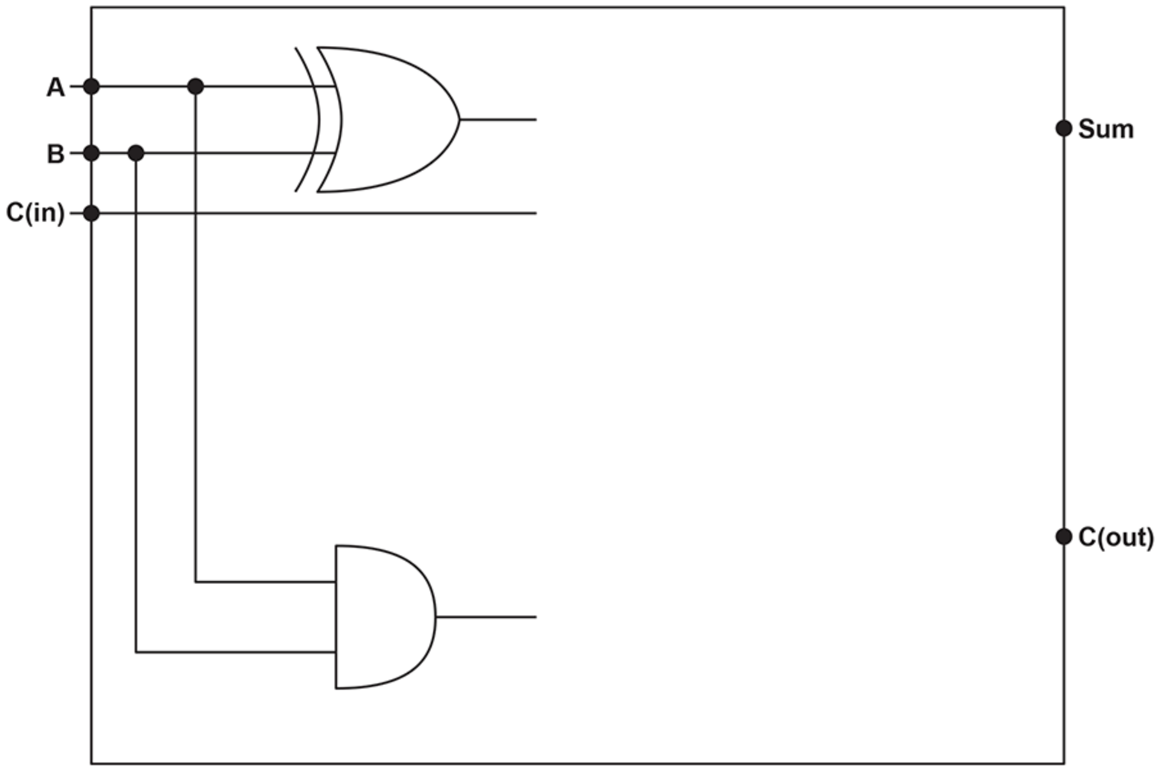
The full adder circuit contains **three** inputs:

- **A** – represents the first input
- **B** – represents the second input
- **C(in)** – represents the carry bit from the previous stage of the addition.

The full adder circuit contains **two** outputs:

- **Sum** – represents the result of adding the three inputs **A**, **B** and **C(in)**
- **C(out)** – represents the carry that is passed to the next stage of the addition.

Complete the logic circuit to represent a full adder.



[4]

(d) This Karnaugh map represents a logic circuit.

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	1	1	1
	01	0	1	1	0
	11	0	0	0	0
	10	0	0	1	1

Use this Karnaugh map to find the simplified expression for this circuit.

You should annotate the Karnaugh map to show the groups that you have used to find the simplified expression.

Write your simplified expression here.

[4]

6(a) DCC Software is a company that creates a wide range of software products for home computers. Its products include image, video and sound editing software.

DCC Software makes significant use of software libraries.

Explain the benefits of using software libraries to create software.

[4]

(b) Once DCC Software has written a piece of software, they use translators to translate the programming code.

i. The table shows different stages of code compilation labelled **A**, **B**, **C** and **D**.

A	Lexical analysis
B	Syntax analysis
C	Code generation
D	Code optimisation

In each box write the letter of the stage that matches each description.

Converts the high level code into low level instructions ☐

Examines the code and improves the efficiency of it without affecting the functionality ☐

Removes comments and whitespace and then converts the high level code into tokens ☐

Identifies syntax errors and creates an abstract syntax tree ☐

[4]

ii. DCC Software is considering the benefits of both an interpreter and a compiler.

State **two** benefits of using an interpreter.

Benefit 1 ____

____

Benefit 2 ____

____ **[2]**

iii. State **two** benefits of using a compiler.

Benefit 1 ____

____

Benefit 2 ____

____ **[2]**

(c) DCC software uses a database table called `TblSoftware` to store details about all of their products.

ProductID	Name	Type	Price
`DCC01`	`DCCPhotos`	`Image`	`£129.99`
`DCC02`	`DCCDraw`	`Image`	`£119.99`
`DCC03`	`DCCAnimate`	`Image`	`£99.99`
`DCC04`	`DCCMovie`	`Video`	`£189.99`
`DCC05`	`DCCEffects`	`Video`	`£99.99`
`DCC06`	`DCCAudio`	`Sound`	`£119.99`

`TblSoftware`

i. Write an SQL statement to insert this new product into `TblSoftware:`

ProductID:	`DCC07`
Name:	`DCCCast`
Type:	`Sound`

Price:	£89.99

[3]

ii. Write an SQL statement to return the `Name` of products that have a `Price` that is less than the `Price` of the product with the `Name` “DCCDraw”.

You should use a nested SELECT statement in your answer.

[5]

7(a) OCR Photo is a free website designed to use Artificial Intelligence (AI) to generate unique photographs from a text description given by the user.

The AI learns by searching the internet randomly for millions of photographs and then matching these to a textual description of the photograph. The photographs include people, property, landscapes and objects.

Discuss the legal and moral implications of this approach and what OCR Photo should consider.

You should include the following in your answer:

- what is meant by AI
- the legal and moral implications of the learning approach described in the question
- a conclusion justifying if OCR Photo should use this approach or not.

[9]

(b) OCR Photo is a free website designed to use Artificial Intelligence (AI) to generate unique photographs from a text description given by the user.

OCR Photo uses run length encoding to compress the generated photographs.

A black and white photograph is encoded using **B** for the colour black and **W** for the colour white. The encoded sequence is:

```
W4B1W4
W3B3W3
W2B5W2
W1B7W1
```

Use the grid to show the result of using run length encoding to decompress this sequence.

[4]

8(a) Orla is developing a program that will allow candidates to complete quizzes for a range of subjects. Each quiz has **three** questions and each one is multiple choice with **three** possible answers.

Orla is considering using a flat file database to store data about candidates, and the subjects and quizzes they sit.

Explain the limitations of using flat file databases.

[4]

(b) Due to the limitations of flat file databases, Orla decides to create a relational database containing these **four** tables:

- `TblCandidate` – Stores data about the candidates
- `TblSubject` – Stores data about the subject
- `TblQuiz` – Stores the questions for each individual quiz and the answers
- `TblResponse` – Stores the candidate's responses for a given quiz.

The fields in each table are shown here. **PK** indicates the primary keys and **FK** indicates the foreign keys that are used.

TblCandidate
CandidateID(PK)
CandidateName
eMail
PhoneNumber

TblSubject
SubjectID(PK)
SubjectName
Description

TblQuiz
QuizID(PK)
Q1Text
Q1Answer
Q2Text
Q2Answer
Q3Text
Q3Answer
SubjectID(FK)

TblResponse
ResponseID(PK)
Date
Q1Response
Q2Response
Q3Response
Score
QuizID(FK)
CandidateID(FK)

Complete an Entity Relationship Diagram (ERD) to show the relationships between these four tables.

TblCandidate   TblSubject

TblResponse   TblQuiz

[3]

(c)

i. State **two** requirements for a database to be in Second Normal Form (2NF).

1 ____

____

2 ____

____

____ **[2]**

ii. State **two** requirements for a database to be in Third Normal Form (3NF).

1 ____

____

2 ____

____

____ **[2]**

(d) Orla wants a web based version of the quiz. She develops a function called `checkAnswers` to calculate the score of each quiz on the client side using JavaScript.

Next to each candidate answer is an empty HTML element. When the user presses submit, the function, `checkAnswers`, will check if each answer is correct. If so, it will add the word "Correct" into the empty HTML element.

The function, `checkAnswers` takes **three** parameters:

- `responses` – an array of the IDs of the HTML elements that contains the user's responses.
- `feedbacks` – an array of the IDs of the HTML elements that the text "Correct" is to be written to if the candidate gets the question right. The array will be the same length as `responses`.
- `answers` – an array of the correct answer for each of the questions. The array will be the same length as `responses`.

The function returns the final score out of three for the quiz. It loops through each question to calculate if the candidate got it correct and processes it appropriately if they did.

Complete the function `checkAnswers`, using pseudocode or program code.

```
function checkAnswers(responses, feedbacks, ..................................)
{
  var ............................ = 0;
  for(var qCount = 0; qCount < responses.length; qCount++)
  {
    response = document.getElementById(responses[qCount]).innerHTML;
    answer = answers[..................................];
    feedbackElement = document.getElementById(feedbacks[qCount]);
    if (response == answer)
    {
      score = .............................
      feedbackElement.innerHTML = "Correct";
    }
    }
  ...................................... score;
}
```

[5]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1	a		1 mark max for each correct bus to max 3. Data bus ◦ Transmits /carries <u>data/ instructions</u> (between the hardware components / between CPU and memory). Address bus ◦ Transmits /carries the (memory) <u>address /location</u> (where data will be sent to /retrieved from). Control bus ◦ Transmits /carries (control) <u>signal(s)</u> (to control /coordinate CPU operations).	3	<u>Examiner's Comments</u> This question was generally successfully responded to with most candidates given 2 or 3 marks. There were some candidates who misunderstood that a bus is for the transfer of data /addresses /signals and does not have control over components.
	b	i	1 mark for input device with a suitable example and 1 mark for output device with suitable example to max 4 e.g.: <u>Input Device</u> • Keyboard … • …Allow him to enter the game instructions / test the game • Mouse… • …Allow him to select /manipulate objects in graphics programs • Graphics tablet… • …Allow him to draw the characters in his game • Microphone… • …Allows him to record game narration/ instructions <u>Output Device</u> • Monitor… • …Display images / graphics he is working on for the game • Speakers… • …Allows him to edit /hear sound effects in the game	4	The devices and examples must be relevant to something Charlie could be doing when doing computer game design /editing /testing. If device name is wrong, ignore example. If device name given in example space (and no device name has been stated) allow both marks. <u>Examiner's Comments</u> For this question candidates were given a context of designing and editing games. Those candidates that gave an example use related to this context were given full marks. Those candidates who gave less successful responses to this question, tended to give correct input and output devices but generic uses which were not related to the context in the question stem.

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td>ii</td><td>1 mark for identifying a secondary storage type and 1 mark for a suitable reason to max 2.

Type: Solid State (1)
Possible Reasons: e.g.:

• Durable /robust (1)
• Fast read /write speeds (1)
• Large storage capacity (1)

Type: Magnetic (1)
Possible Reasons: e.g.:

• Cheap to buy per unit (1)
• Large (enough) storage capacity (1)
• Fast read /write speeds (1)</td><td>2</td><td>Do not allow optical storage.

The reason must be related to the type of secondary storage provided.

Allow FT to MP2 if suitable device itself is named e.g. HDD drive, flash drive.

<u>Examiner's Comments</u>

This question was generally well responded to with most candidates given full marks. Candidates who lost a mark on this question by giving a device name rather than a type were able to achieve a follow through mark for then giving a valid reason for the device they had given.</td></tr>
<tr><td></td><td>c</td><td>i</td><td>1 mark for each correct row.

<table><tr><th>Statement</th><th>CPU</th><th>GPU</th></tr><tr><td>Suited to performing simple operations in parallel on larger data sets.</td><td></td><td>✓</td></tr><tr><td>Suited to complex operations on smaller data sets.</td><td>✓</td><td></td></tr></table></td><td>2</td><td>Ignore rows with a tick in both boxes.

Allow alternative marks to a tick e.g. 'yes'

<u>Examiner's Comments</u>

Most candidates were able to access both marks for this question. Less successful responses were the wrong way round so were given no marks.</td></tr>
<tr><td></td><td></td><td>ii</td><td>1 mark for each to max 2, e.g.:

• More responsive gameplay
• Shorter loading times /rendering
• More realistic /immersive graphics
• Higher frame rates /smoother gameplay
• Higher resolutions /HD /4k graphics
• Better multiplayer experience</td><td>2</td><td>Must be relevant to graphics in gameplay

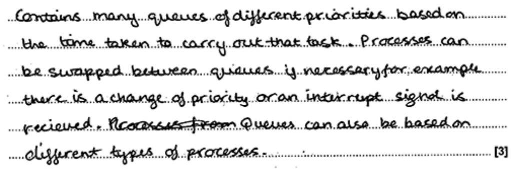
Ignore general points about a GPU /multiple cores or reverse points about benefit for CPU

Only mark first benefit given per answer space

<u>Examiner's Comments</u>

This question asked for candidates to give two benefits which processing and displaying graphics. Many candidates wrote generic responses which gave benefits of a GPU but did not respond to the question being asked.</td></tr>
<tr><td></td><td>d</td><td>i</td><td>• Multi-Tasking</td><td>1</td><td><u>Examiner's Comments</u>

Most candidates were able to access this mark.</td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
		ii	1 mark for each to max 3, e.g.: • Puts **processes** into different categories/ queues • Each queue has a different priority /quantum /scheduling approach • **Processes** can move **between queues…** • …based on their behaviour /priority • Designed to minimise **I**/O bottle necks • …balances longer and shorter processes	3	<u>**Examiner's Comments**</u> Few candidates gained full marks on this question. Many discussed a single queue and missed the point of a multi-level feedback queues (MLFQ) using multiple queues. Exemplar 1  Contains many queues of different priorities based on the time taken to carry out that task. Processes can be swapped between queues if necessary for example there is a change of priority or an interrupt signal is recieved. ~~Processes from~~ Queues can also be based on different types of processes. [3] The candidate has shown a good understanding of multi-level feedback queues. They have shown they understand that there are multiple queues and that these have different priorities. They have shown they understand processes being able to move between queues based on their behaviour /priority and that processes are put into different queues. This candidate was able to gain full marks.
		iii	2 marks for valid explanation. • Allows the **operating-system /software** to communicate /interface with external hardware • Translates operating-system instructions into a form that devices can use • Allows peripherals from different manufacturers to work with different OS • Abstracts hardware complexity (the OS and software don't need to know specific commands)	2	<u>**Examiner's Comments**</u> Most candidates were able to access the mark for allowing a device to communicate with the operating system (OS), but few went on to then give a second point of why /how.

Question			Answer/Indicative content	Marks	Guidance
	e		1 mark for suitable utility software and 1 mark for purpose to max 4, e.g.: • **Defragmentation…** • …puts fragmented files in consecutive storage locations • **Encryption…** • …scrambles data so it cannot be understood • **Compression…** • …reduces the amount of storage space files take up /creates more storage space • **File repair…** • …repairs damaged /corrupt files • **Backup…** • …copies data to another storage device • **Security /anti-virus…** • …protects the computer against threats / malware /hacking etc. • **System clean-up…** • …removes files that are no longer needed	4	Accept firewall, file management, disk management with reasonable purpose <u>Examiner's Comments</u> Most candidates could state two utilities however some candidates gave less successful responses by giving a vague purpose. There were a few candidates who responded with applications rather than utilities.
			Total	23	

Question			Answer/Indicative content	Marks	Guidance
2	a	i	2 marks for each difference to max 4. • **Von Neumann** stores data and instructions in the same memory unit • **Harvard** stores data and instructions in separate memory units • **Von Neumann** fetches data and instructions sequentially /follows FDE cycle • **Harvard** can fetch data and instructions at the same time • **Von Neumann** uses a single bus for both data and instructions • **Harvard** uses different buses for data and instructions	4	Must be a full comparison for 2 marks to be given. Each comparison must be comparing the same feature. **Examiner's Comments** This question has been a common question on previous question papers and as such was well responded to by many candidates. There are still some candidates who did not use the correct terminology which would be expected in a question of this type. Exemplar 2 Difference 1 Von Neumann stores program data and instructions in the same area in memory. Harvard stores program data and instructions in different memory areas. Difference 2 Von Neumann uses the same bus to transfer program data and instructions. Harvard uses different buses for transferring data and instructions. [4] The candidate has used correct terminology and has given a clear comparison covering both Von Neumann and Harvard architecture. **Assessment for learning - Comparison questions** Candidates should be encouraged to give a full comparison of both sides in comparison questions. A candidate giving the response as' Von Neumann stores data and instructions in the same memory unit and Harvard does not' will only be able to access the first mark.
		ii	1 mark for each to max 2. • Makes the CPU more efficient • Increases the throughput of the CPU • Allows parts of the FDE cycle to happen simultaneously (so increases performance) • Reduces latency/ idle time	2	DNA **'faster'/'more efficient'** on their own. **Examiner's Comments** Generally, this question was well responded to with the biggest misconception being to state multiple cores or parallel processing.

Question			Answer/Indicative content	Marks	Guidance
		iii	2 marks for each difference to max 4. • **CISC** has a larger instruction set • **RISC** has a smaller instruction set • **CISC** requires more complex hardware/ requires cooling • **RISC** requires less complex hardware/ requires little cooling • **CISC** takes multiple **clock** cycles to execute an instruction • **RISC** takes one **clock** cycle to execute an instruction • **CISC** tends to use more energy • **RISC** tends to use less energy • **CISC** has more addressing modes • **RISC** has fewer addressing modes • **CISC** uses less RAM • **RISC** uses more RAM • **CISC** is difficult to pipeline • **RISC** is easier to pipeline	4	Must be a full comparison for 2 marks to be given. Each comparison must be comparing the same feature. <u>Examiner's Comments</u> As in all comparison questions candidates need to make sure that they give a full comparison and do not just list unrelated features of each. Most candidates were able to access marks on this question which has been a question on previous question papers.
		iv	1 mark for each to max 2. • Multiple **instructions /threads** can be processed at the **same time** • Improves multi-tasking • Enables parallel processing • Increases the performance of the computer	2	Allow 'FDE Cycles' for MP1 <u>Examiner's Comments</u> Many candidates were able to access the marks for parallel processing and multiple instructions being processed at the same time.
	b		**Mark Band 3 – High level (7-9 marks)** The candidate demonstrates a thorough knowledge and understanding of virtual machines. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence /examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically	9	Answers may include, but are not limited to, some of the points below: **AO1** Virtual Machines: • A machine that exists as software • Behaves as another program or piece of hardware • Needs a real machine to run on • For the user it will be indistinguishable Virtual Storage: • A file will be visible to a virtual machine as secondary storage • Remote storage that is mounted locally to seem like local storage • Stored on remote servers.

Question			Answer/Indicative content	Marks	Guidance

			structured. The information presented is relevant and substantiated.
			
			Mark Band 2 – Mid level (4-6 marks)
			
			The candidate demonstrates reasonable knowledge and understanding of virtual machines. The material is generally accurate but at times underdeveloped.
			
			The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence /examples are for the most part implicitly relevant to the explanation.
			
			The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed.
			
			There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.
			
			Mark Band 1 – Low Level (1-3 marks)
			
			The candidate demonstrates a basic knowledge of some aspects of virtual machines. The material is basic and contains some inaccuracies.
			
			The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided.
			
			The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way.
			
			The information is supported by limited evidence and the relationship to the evidence may not be clear.
			
			0 mark
			
			No attempt to answer the question or response is not worthy of credit.
					Thick /Thin client:
					
					• Thick client involves all the processing being completed on the client. This can result in minimal use for a server, possibly only storing common data
					• Thin client means the client just deals with user input and output which are transmitted to a server. The server does all the complex processing
					
					AO2 Application
					
					Thin client approach:
					
					• Potentially safer to store the database /sensitive data on virtual servers as the underlying system is not vulnerable to security breaches.
					• As the virtual machine is software, if it is compromised it can be terminated and restarted from a safe state
					• More efficient use of hardware as storage /processing resources are shared // not sat idle on a laptop not being fully utilised
					• Can dynamically shift resources
					  ◦ If a user is doing a processor intense task more virtual cores can be allocated to them // a user doing low intensity tasks can have fewer
					  ◦ If a user needs larger storage requirements, space can be dynamically allocated to them
					• There is limited offline functionality and users may only be able to work if there is an internet connection.
					• It's difficult to personalise for individual users.
					• May need a lot of bandwidth to be able to access the large databases and carryout the data processing.
					
					Thick client approach:
					
					• The computers have their own processing power to run complex processing locally.
					• Often has offline functionality as they can continue to work without an internet connection
					• Devices can be customised to meet the user's needs.
					• Possibly less secure (than thin client) as the security software on the devices

Question			Answer/Indicative content	Marks	Guidance

					needs to be managed.
					• May need more advanced devices so therefore will increase the upfront costs.
					
					AO3 Evaluation
					
					• There are cost savings to using thin client approaches.
					• The ability to dynamically allocate resources would benefit users who have unusually large processing jobs as they will not cause unacceptable wait times.
					• As the users are mobile, they are reliant on network access or mobile access.
					  ◦ May not allow 3rd party access.
					  ◦ Mobile data access may be slow or even non-existent.
					• If a user has a fully powered laptop in case of connectivity issues it will remove the cost saving advantages
					• If there is slow network access the user could be subject to unacceptable waits due to latency issues.
					
					Examiner's Comments
					
					Many candidates were able to gain marks on this question across the three mark bands. Some candidates showed good understanding and were able to give a well justified conclusion. There were some candidates who merely repeated the information given in the question stem without any further expansion or application. Some candidates talked about cloud storage and then went on to describe how this would be insecure due to being held by a different company which was not relevant in this scenario. There were some candidates that confused virtual memory and virtual storage and gave descriptions of how virtual memory works.

			Total	21	

Question			Answer/Indicative content	Marks	Guidance
3	a	i	1011 1101	1	<u>Examiner's Comments</u> Question 3 (a) (i) and Question 3 (a) (ii) asked for a simple conversion and in general they were both successfully responded to by candidates.
		ii	BD	1	<u>Examiner's Comments</u> Question 3 (a) (i) and Question 3 (a) (ii) asked for a simple conversion and in general they were both successfully responded to by candidates.
		iii	228	1	<u>Examiner's Comments</u> Many candidates were able to gain the mark here while some gave a response in binary instead of denary and missed the mark.
	b		1 mark for each to max 2 • Divides… • …by 8 / 2^3	2	**Allow**: multiplies by 2^{-3} or ⅛ or 0.125 (1) <u>Examiner's Comments</u> Many candidates were able to access 1 or 2 marks here. Many were aware that there was division but then went on to give a less successful response for the second mark.
	c		1 mark for each to max 6 • Both exponents converted 0011 = 3 0010 = 2 • Both mantissas shifted 0110.10 010.010 • Align mantissas: 0110.10 010.010 • Binary subtraction: 02 0110.~~1~~00 0<u>010.010</u> 0100.010 • Correct mantissa 010001 • Correct exponent 0011	6	Correct answer with valid **binary working for calculation** = full marks. Where denary working is clearly used to calculate the answer, do not allow MP4 <u>Example of 2's complement addition</u> Convert 2^{nd} mantissa to 2's comp e.g.: 010.010 >> 101.11 0110.10 + <u>1101.11</u> <u>+ 0100.01</u> 1111 <u>Examiner's Comments</u> This was an accessible question with many candidates gaining all available marks. Most candidates gained the mark for converting the exponent and moving the point. There were many ways that this question could be responded to, and all relevant methods were given marks.

Question			Answer/Indicative content	Marks	Guidance
	d	i	1 mark for each to max 2 • Nibble 1 - 1010 • Nibble 2 - 0010 Solution: Byte **1010 0011** AND Mask **1010 1110** Result **1010 0010**	2	Examiner's Comments Most candidates were able to access the marks for this question although some added the Byte and the AND mask
		ii	1 mark for: • To extract /enable /disable bits in a value	1	Allow: To encrypt data Examiner's Comments This question was less successfully responded to and many candidates stated what it did rather than stating the purpose of the bitwise and mask.
			Total	14	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>4</td><td>a</td><td></td><td>1 correct answer = 1 mark
2 correct answers = 2 marks
5 correct answers = 3 marks

<table>
<tr><th>Description</th><th>Command</th></tr>
<tr><td>Load the value stored in the memory address into the accumulator.</td><td>LDA / LOAD</td></tr>
<tr><td>Branch without checking the value in the accumulator</td><td>BRA / BR</td></tr>
<tr><td>Store the value from the accumulator into a memory address</td><td>STA / STO</td></tr>
<tr><td>Branch if zero is in the accumulator</td><td>BRZ / BZ</td></tr>
<tr><td>Branch if zero or a positive number is in the accumulator</td><td>BRP / BP</td></tr>
</table>
</td><td>3</td><td><u>Examiner's Comments</u>

Most candidates were able to access all 3 marks with some making an error on one and still gaining 2 marks.</td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance

	b		1 mark for each to max 4:
			
			• Correctly inputting two values and storing at least one value
			• Correctly subtracting one number from the other
			• Correct use of branch
			• Correct output for each path
				4	Accept alternative mnemonics from the spec: STO, LOAD, BR, BZ, BP, IN, INPUT, COB, END.
					
					If labels used must be a corresponding DAT location for MP1.
					
					POSSIBLE SOLUTION:
					
					`             INP`
					`             STA   FIRST`
					`             INP`
					`             STA   SECOND`
					`             SUB   FIRST`
					`             BRP   OUTPUTSECOND`
					`             LDA   FIRST`
					`             OUT`
					`             HLT`
					`OUTPUTSECOND LDA   SECOND`
					`             OUT`
					`             HLT`
					`FIRST        DAT   0`
					`SECOND       DAT   0`
					
					<u>Examiner's Comments</u>
					
					There were varied responses to this question. Many candidates clearly had access to LMC and had practised writing programs in it and were able to gain full marks. Some candidates incorrectly output the correct number but then did not stop the program, so it then went on to output the second number as well.

Question			Answer/Indicative content	Marks	Guidance
	c		**Mark Band 3 – High level (9-12 marks)** The candidate demonstrates a thorough knowledge and understanding of high /low level languages. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence /examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. **Mark Band 2 – Mid level (5-8 marks)** The candidate demonstrates reasonable knowledge and understanding of high /low level languages. The material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence /examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. **Mark Band 1 – Low Level (1-4 marks)** The candidate demonstrates a basic knowledge of some aspects of high /low level languages. The material is basic and contains some inaccuracies.	12	Answers may include, but are not limited to, some of the points below: **AO1 Knowledge** High Level Languages: • Uses English-like syntax /words. • Easier for humans to read, create and debug • Is problem orientated • Not locked to any particular architecture (but needs a translator for the given architecture) • Programmer is not in control of what machine code instructions are generated by the translator. Low level languages: • Written in binary / 1s and 0s. • Processor specific. • Programs are based around the instructions available. • Programmer has direct control over what instructions are being run and what memory is being used. • Provides an interface between programmer and hardware. **AO2 Application** 1. This program could be written in a **high level language**. This is because it will most likely contain a user interface that makes various network requests to fetch the news and then present this to the user. 2. This program could be written in a **low level language** as it is for an embedded system it may have a novel architecture and have no translator. It would also need direct access to the hardware and memory management. 3. This program could be written in a **high level language**. This is because it will contain a user interface that will most likely require a lot of functionality which can be simplified in a high level language. 4. This program could be written in **either**. Low level would give direct access to the memory management to better handle large amounts of data being processed. Higher level may

Question			Answer/Indicative content	Marks	Guidance

			The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided.
			
			The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way.
			
			The information is supported by limited evidence and the relationship to the evidence may not be clear.
			
			0 mark
			No attempt to answer the question or response is not worthy of credit.
					have libraries for the data manipulation required.
					5. This program could be written in a **low level language**. This is because it will need to have direct access to the computer hardware to deal with memory allocation /deallocation, buffers, data structures, registers, interrupts etc.
					
					AO3 Evaluation
					
					• Low level languages allow you to direct access the devices resources such as the memory to allow you to optimise the devices performance.
					• Low level languages are commonly used in embedded systems as they have limited resources available.
					• Low level languages are commonly used in real-time systems when speed is a crucial factor
					  ◦ Direct control of instructions means the ability to control the speed
					  ◦ Direct control over the memory addresses used allows control of the memory footprint
					• High level languages allow for faster development. They contain built-in functions to write code quicker.
					• High level languages are work across different platforms so they can be run on different operating systems.
					
					<u>Examiner's Comments</u>
					
					There were a range of responses to this question across the three mark bands. Candidates who accessed the top mark band gave a good description of the difference between high and low level programming languages (PLs); they were able to give good justification for each project choice and gave a conclusion that justified why both were needed. Those who accessed the middle mark band gave a reasonable description of both and were able to apply their knowledge to the five projects but either gave a weak conclusion or were unable to give enough detail to explain why high or low level would be suitable for each project. Those who accessed the low mark band had basic knowledge of what a high level and a low level programming language are and

Question			Answer/Indicative content	Marks	Guidance
					tended to say high level was for complex problems and low level was for easy problems which did not require much code. **Assessment for learning** Some candidates confused what low level programming language (PL) is and either said it was binary and needed no translation or that procedural languages were low and Object- Oriented (OO) were high. The other common misconception was that low level is just for simple programs while high level is for more complex programs. Exemplar 3 A high-level programming language is a language that is closer to the English language than it is to Assembly/machine code. It is the modern Standard/norm of programming and software development. A low-level language is a language that ~~employees~~ employs shorter and more general syntax, like Assembly, which uses mnemonics, and is further in similarity to English. Low-level languages typically require more specialised programmers. Project 1 and 3 ~~may~~ will be suitable for development with a higher level programming language as the projects ~~employ~~ demand the need for complex features, like developing the user-interface for both project 1 or 2 or ~~implementing http requests using web~~ sending web requests in Project1. Features like the user interface and web requests are typically done with the use of pre-existing libraries. © OCR 2025

Question			Answer/Indicative content	Marks	Guidance
					that are not ~~accessible on usable on~~ easily usable on low-level languages, and ~~could result in Koti~~ choosing a low-level programming language ~~for that~~ these projects could result in Koti investing significantly more time than necessary on implementing these ~~the~~ features Project 2 and 5 are most suitable for development with a low-level programming language, as both projects involve interfacing with hardware. ~~which~~ Project 2 benefits ~~from~~ low-level language ~~usage as~~ usage as software running on the embedded device will ~~the~~ need to compact in file size, ~~and~~ which low-level languages prove to be ~~better in~~ lower in file size. Project ~~4~~ 5 also ~~requires~~ benefits from the use of a low level language ~~as~~ as device drivers need to be very optimised ~~and~~ 4 c as ~~the~~ to prevent delays in signals processing between ~~device and~~ hardware and software. Low-level languages provide more ~~a~~ control over ~~sys~~ the system (like memory management) which promotes effective optimization Project 4 stands to benefit from low-level languages as the computation of ~~the~~ such large amounts of data is unlikely to require many libraries, ~~and also~~ but also require lots of optimization and ~~improved~~ execution speed, which low-level languages excel in. In conclusion, these different languages are needed as high-level languages promote quicker development and more robust solutions of complex applications through the use of libraries and heavily abstracted readable syntax, and low-level languages provide the developers with increased control over the system, such as memory, which makes low-level languages more suitable for optimized solutions that require lots of computation ~~the~~ or software that interacts with hardware The candidate has given a successful description of both high and low level programming languages. They have given relevant justification for each choice and have shown good understanding of the requirements of each project. They have given a successful conclusion.
			Total	19	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>5</td><td>a</td><td></td><td>1 mark for each correct row.

<table>
<tr><th>A</th><th>B</th><th>S</th><th>C</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td></tr>
</table></td><td>4</td><td><u>Examiner's Comments</u>

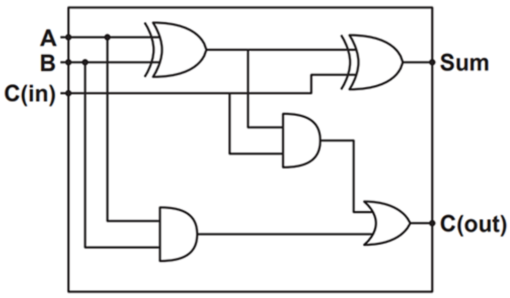
Generally, this question was successfully responded to with most candidates accessing full marks</td></tr>
<tr><td></td><td>b</td><td></td><td>1 mark for each to max 2:

• A half adder can only add 2 single bits // has no third bit
• Full adder can handle a 3rd bit / carry in</td><td>2</td><td>Allow: 'has no carry bit' for MP1

<u>Examiner's Comments</u>

This question was successfully responded to although some candidates gave a description of the given diagram rather than a difference.</td></tr>
<tr><td></td><td>c</td><td></td><td>1 mark for each to max 4:

• New XOR joining output of first XOR and Cin
• New AND joining output of first XOR and Cin
• New OR joining both AND gates
• Sum and Cout connected, all correct and no additional gates

Solution:

</td><td>4</td><td><u>Examiner's Comments</u>

Most candidates were able to access 1 mark but few could complete the logic circuit and access full marks. Less successful responses showed the first AND being an input for the next AND.</td></tr>
</table>

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td>d</td><td></td><td>1 mark for each to max 4:

• Correctly identifying group 1 (red group) and correctly identifying group 2 (blue group) using wrap around
• Identifying (B∧¬C) / B AND NOT C
• Identifying (A∧¬D) / A AND NOT D
• Joining all expressions with an OR // v // +

Solution: (A∧¬D) ∨ (B∧¬C)
<table>
<tr><td></td><td></td><td colspan="4">AB</td></tr>
<tr><td></td><td></td><td>00</td><td>01</td><td>11</td><td>10</td></tr>
<tr><td rowspan="4">CD</td><td>00</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>01</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>11</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>10</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
</table></td><td>4</td><td>Accept alternative notation:
AND // ^ // •
OR // v // +
NOT // ¬ // ‾‾

DNA incorrect ¬ shape

Allow either order e.g.: B^¬C // ¬C^B

<u>Examiner's Comments</u>

Most candidates were able to access some marks on this question with many gaining full marks. As is usual with this type of question, alternative notation was given mark.</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>14</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
6	a		1 mark for each to max 4. • Reduces **development /debugging** time. • Reduced programming knowledge is needed. • Libraries will already be tested. • Libraries are often optimised for performance. • Libraries are designed to work across multiple platforms. • Consistency between applications • Reduces file sizes (DLLs)	4	Allow: 'saves developers time' DNA 'saves time' on its own <u>Examiner's Comments</u> Most candidates were able to explain that libraries have been pre tested and save on development time with some being also saying libraries are optimised. Few candidates were able to access full marks, but many were given 2 or 3 marks.
	b	i	1 mark for each correct match to max 4. C D A B	4	<u>Examiner's Comments</u> Generally, this question was well responded to with many candidates accessing all 4 marks.
		ii	1 mark for each to max 2. • Easier /faster to debug /fix errors /maintain • Easier /faster to test code **line by line** • Code will still run until the first error • Code is portable across different architectures	2	<u>Examiner's Comments</u> Both questions generally successfully responded to however some candidates gave a feature without saying what the benefit was.
		iii	1 mark for each to max 2. • Do not need to distribute <u>source</u> code // protects intellectual property • Do not need run-time environment // code is not re-translated each time it runs • Faster run-time (once compiled) • Produces an executable file • Code is <u>optimised</u>	2	<u>Examiner's Comments</u> Both questions generally successfully responded to however some candidates gave a feature without saying what the benefit was.

Question			Answer/Indicative content	Marks	Guidance
	c	i	1 mark for each to max 3. • `INSERT INTO TblSoftware (ProductID, Name, Type, Price)` • `VALUES` • `(“DCC07”, “DCCCast”, “Sound”, 89.99)`	3	Field and table names must be in the correct case as given. MP1 does **not** require field names. Allow Price in speech marks e.g.: “89.99” or “£89.99”. For full marks a fully correct working answer must be provided. Max 2 if the order of MP1/2/3 is not correct. <u>Examiner’s Comments</u> Few candidates gained full marks on this question. Insert is one of the SQL commands in the specification and candidates should be given the opportunity to practise with the full range of the SQL commands included in appendix 5d of the specification.
		ii	1 mark for each correct line to max 5. • `SELECT Name FROM TblSoftware` • `WHERE Price <` **(** • `SELECT Price` • `FROM TblSoftware` • `WHERE Name = ‘DCCDraw’`**)**	5	Field and table names must be in the correct case as given. Allow FT for missing brackets. Allow == for comparison. <u>Examiner’s Comments</u> Many candidates gained full marks on this question with many being able to gain some marks.
			Total	20	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>7</td><td>a</td><td></td><td>Mark Band 3 – High level (7-9 marks)

The candidate demonstrates a thorough knowledge and understanding of the moral and legal implications of AI. The material is generally accurate and detailed.

The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence /examples will be explicitly relevant to the explanation.

The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered.

There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.

Mark Band 2 – Mid level (4-6 marks)
The candidate demonstrates reasonable knowledge and understanding of the moral and legal implications of AI. The material is generally accurate but at times underdeveloped.

The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence /examples are for the most part implicitly relevant to the explanation.

The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed.

There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.

Mark Band 1 – Low Level (1-3 marks)

The candidate demonstrates a basic knowledge of some aspects of the moral or legal implications of AI. The material is basic and contains some inaccuracies.</td><td>9</td><td>Answers may include, but are not limited to, some of the points below:

AO1 Knowledge
AI behaves and performs tasks that normally require human intelligence. AI needs to learn by understanding languages, recognising patterns, solving problems and learning from experience. They can be trained to complete specific tasks or can be used to complete a wide range of different tasks.

AO2 Application
Moral

• Individuals may not know that their images are being used to train the AI.
• Individuals are entitled to privacy and they may not want their photograph to be used by the AI. They may think that this is an invasion of privacy.
• The AI is searching the internet randomly and therefore, it needs to ensure that it uses a complete range of diverse photographs such as people from different cultures and diversity of scenes to represent reality and ensure fairness.

Legal

• It needs to ensure that the photographs are accessed legally without criminal activity to ensure this does not breach the Computer Misuse Act.
• Organisations must ensure that they keep personal data safe under the Data Protection Act. This includes photos and therefore they need to ensure these cannot be accessed by the AI.
• Some images will be protected by the Copyright, Design and Patents Act which includes people's intellectual property. These therefore cannot be accessed without permission from the author.

A03 Evaluation

• AI provides many benefits, however, for it to be effective it needs to learn and be trained.</td></tr>
</table>

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td></td><td>The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided.

The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way.

The information is supported by limited evidence and the relationship to the evidence may not be clear.

0 mark No attempt to answer the question or response is not worthy of credit.</td><td></td><td>• Learning from paired photos and descriptions is a good approach to use to allow the AI to learn, however they should consider:
◦ ensuring that all relevant computer laws and other non-computer laws are followed.
◦ only using royalty free images to avoid copyright infringements.
◦ ensuring individuals have given their consent before their photographs are used.
◦ ensuring that a wide range of photographs are used to ensure the AI is not biassed.

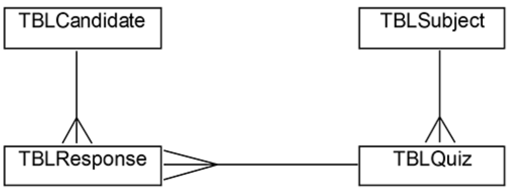
<u>Examiner’s Comments</u>

There were a range of responses to this question with many candidates being able to give both legal and moral implications related to the scenario given. Some had clearly used past question papers to revise and were able to talk about possible bias which was relevant to the scenario. Most candidates mentioned ‘The Copyright Design and Patents Act’ and some were able to discuss the difficulties surrounding this with some giving relevant current legal reform and discussion of the debates currently happening which was pleasing to see.</td></tr>
<tr><td></td><td>b</td><td></td><td>1 mark for each correct row to max 4.

<table><tr><td>W</td><td>W</td><td>W</td><td>W</td><td>B</td><td>W</td><td>W</td><td>W</td><td>W</td></tr><tr><td>W</td><td>W</td><td>W</td><td>B</td><td>B</td><td>B</td><td>W</td><td>W</td><td>W</td></tr><tr><td>W</td><td>W</td><td>B</td><td>B</td><td>B</td><td>B</td><td>B</td><td>W</td><td>W</td></tr><tr><td>W</td><td>B</td><td>B</td><td>B</td><td>B</td><td>B</td><td>B</td><td>B</td><td>W</td></tr></table></td><td>4</td><td>Allow shading, or letters or words etc to show understanding of each run.

<u>Examiner’s Comments</u>

Most candidates responded successfully to this question and gained full marks.</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>13</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
8	a		1 mark each to max 4. • Lots of repeated / redundant data • Can lead to data being inconsistent • Increases the size of the database • Data access speed is slow • Searches are slow /hard • They are difficult to manage /maintain • It is difficult to add additional functionality • Difficult to apply ACID principles	4	DNA 'information' for 'data' – penalise once and then FT. **Examiner's Comments** Most candidates were able to access at least 2 marks for this question. Few gave enough points to be given full marks.
	b		1 mark for each correct relationship: • One to many TblCandidate to TblResponse • One to Many TblQuiz to TblResponse • One to Many TblSubject to TblQuiz 	3	MAX 2 if any extra relationships added. DNA: words should not be used to describe relationships. Accept: relationship symbol from specification only: One-To-Many relationship  **Examiner's Comments** Many candidates found this question difficult and there were a range of less successful responses ranging from tables being joined with 1-1 relationships to all tables being joined in many to many relationships. Candidates should be encouraged to look at the keys in the information given and the relationship between a primary key and a foreign key.
	c	i	1 mark each to max 2 • Data is in 1st normal form • All non-key fields depend on key field(s) /no partial dependencies	2	**Examiner's Comments** Most candidates were aware that the database needed to be in first normal form for it to be in second normal form and that it needed to be in second normal form for it to be in third normal form. Fewer were able to then give a correct second point with some discussing atomicity and no repetition. This type of question has been in previous question papers and candidates should use those to practise.

Question			Answer/Indicative content	Marks	Guidance
		ii	1 mark each to max 2 • Data is in 2nd normal form • All non-key fields don't depend on another non-key field/ no transitive dependencies	2	<u>Examiner's Comments</u> Most candidates were aware that the database needed to be in first normal form for it to be in second normal form and that it needed to be in second normal form for it to be in third normal form. Fewer were able to then give a correct second point with some discussing atomicity and no repetition. This type of question has been in previous question papers and candidates should use those to practise.

	d		1 mark for each missing statement to max 5.
			
			• `answers`
			• `score`
			• `qCount`
			• `score + 1`
			• `return`
			
			`function checkAnswers(responses, feedbacks, answers)`
			`{`
			`   var score = 0;`
			`   for(var qCount = 0; qCount < responses.length; qCount++)`
			`   {`
			`      response = document.getElementById(responses[qCount]).innerHTML;`
			`      answer = answers[qCount];`
			`      feedbackElement = document.getElementById(feedbacks[qCount]);`
			`      if (response == answer)`
			`      {`
			`         score = score + 1`
			`feedbackElement.innerHTML = "Correct";`
			`      }`
			`      }`
			`      return score;`
			`}`
				5	Allow: for MP5 return via function call e.g. `checkAnswers=`
					
					<u>Examiner's Comments</u>
					
					This question was well responded to by many candidates.

			Total	16	